

Prophylactic Radiation Therapy for High-Risk Asymptomatic Bone Metastases: A New Standard of Care or Need for More Data?

TO THE EDITOR:

We read with interest the recent publication by Gillespie et al¹ on prophylactic radiation therapy (RT) for high-risk asymptomatic bone metastases. This multicenter, randomized phase II trial demonstrated that prophylactic RT could reduce skeletal related events (SREs) and hospitalizations and was associated with improved overall survival (OS). The authors are to be congratulated for conducting this trial that addresses a clinically relevant question in radiation oncology. However, there are several issues that are worth considering when interpreting their results.

First, the mechanism of prophylactic RT on OS remains unclear. It is unknown whether some of the treated bone metastases might represent oligoprogressive disease and that these cases drove the improvement in the OS. More patients in the prophylactic RT arm had Karnofsky Performance Status of 90 or above when compared with the observation arm (62% v 46%). There were fewer bulky bone metastases in the prophylactic arm when compared with the observation arm (16% v 24%), which could be associated with higher fracture risk. Validated instruments for baseline fracture risk, such as the Mirel's Scoring System or the Spinal Instability Neoplastic score for long bones and spine, respectively, were not used in this study.^{2,3} In addition, the inclusion of reirradiation or subsequent RT as SRE is prone to bias as the decision of whether to give radiotherapy can be subjective.⁴ The type and line of systemic treatment and the proportion of patients with visceral metastases in each arm were not reported. These differences might further explain some of the apparent benefits in the prophylactic RT arm. A future phase III trial, as suggested by Gillespie et al,¹ should be designed to ensure that the baseline risks are comparable in both arms.

Second, various dose-fractionation schedules, including both stereotactic body radiation therapy (SBRT) and conventional external beam radiotherapy (cEBRT) with a single 8 Gy fraction, 20 Gy in five fractions, and 30 Gy in 10 fractions, were used in this trial. There is a complex interaction between the dose-fractionation and SREs. If SREs are related to progression of bone metastases, theoretically a higher local control may better protect against SREs. Spine SBRT has been reported to yield improved local control compared with cEBRT.⁵ However, current guidelines do not recommend SBRT as standard of care for complicated bone metastases, in part, because of its higher risk of SBRT-induced fracture.⁶ There is some evidence to suggest that for cEBRT, a longer course of radiotherapy may lead to better local control and more remineralization of lytic bone metastasis.^{7,8} The

heterogeneous radiotherapy dose-fractionation schedules used in this trial do not allow us to draw firm conclusion on the optimal dose-fractionation schedule to use in the prophylactic setting. A future phase III trial using a standardized dose-fractionation schedule is warranted.

Third, all bone metastases in this study were detected by cross-sectional imaging; however, the specific imaging modality used was not reported. In late-stage castration-resistant prostate cancer, the UK PROMPTS trial found that the use of screening spinal magnetic resonance imaging (MRI) and pre-emptive treatment to prevent spinal cord compression did not improve the time to clinical spinal cord compression.⁹ This suggests that some patients would not benefit from screening and subsequent RT for asymptomatic bone metastases. Positron emission tomography-computer tomography (PET-CT) could be more sensitive than CT in detecting bone metastases in breast cancer, and it is not clear whether bone metastases visible only on PET-CT should be treated similarly to those detected by CT or MRI.¹⁰ A future phase III trial will need to choose the optimal type of imaging modality for diagnosing the asymptomatic bone metastases that balances resources and practicability.

We applaud the investigators for conducting this study and fully agree with them that the findings have to be confirmed in a larger phase III clinical trial. Some of the end points in this phase II study reached statistical significance, but given its sample size of only 78 patients, it should best be regarded as hypothesis generating and not practice changing. The future phase III trial needs to address the issues of patient selection, risk of fracture assessment, dose-fractionation schedule of RT, and the imaging modality used to detect asymptomatic bone metastases.

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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